

Consumer Choice

Alejandro Melo Ponce



Stony Brook University

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Indifference curves and the marginal rate of substitution

- Consider the case $X = \mathbb{R}_+^2$. and assume that preferences are represented by u .
- Indifference curve U_1 is $\{(x, y) | u(x, y) = U_1\}$
- The equation $U_1 = u(x, y)$ defines, *implicitly* y as a function of x .
- If u is differentiable, we can think about calculating what is the slope of the indifference curve.
- This slope has a very important economic meaning:

Definition

The **marginal rate of substitution** is the *negative* of slope of an indifference curve U_1 at some point. That is

$$MRS = -\frac{dy}{dx} \Big|_{u=U_1} \quad (1)$$

the MRS tells us about the trades that an individual is willing to make, it is the opportunity cost in terms of y of consuming one unit more of x

The MRS: formal derivation

- Suppose that $u(x, y)$ is differentiable.
- An indifference curve: $U_1 = u(x, y)$
- Totally differentiate both sides with respect to x and y

$$dU_1 = u_x dx + u_y dy$$

$$0 = u_x dx + u_y dy \quad \text{Because } U_1 \text{ is a constant}$$

$$\frac{dy}{dx} = -\frac{u_x}{u_y} \Rightarrow MRS = \frac{u_x}{u_y}$$

Definition (Convex Preferences in terms of MRS)

Suppose the utility function is differentiable. Then preferences are convex if the utility function that represents them has decreasing MRS.

- What condition should hold on the second order partial derivatives $(u_{xx}, u_{yy}, u_{xy}, u_{yx})$ that will guarantee a decreasing MRS? More on this later. . .

Utility functions for specific preferences

- Cobb Douglas $u(x, y) = x^\alpha y^\beta$.
- Perfect Substitutes $u(x, y) = \alpha x + \beta y$
- Perfect complements $u(x, y) = \min(\alpha x, \beta y)$
- CES utility: $u(x, y) = \frac{x^\delta}{\delta} = \frac{y^\delta}{\delta}$, when $\delta \leq 1, \delta \neq 0$ and
 $u(x, y) = \ln x + \ln y$ when $\delta = 0$
- Quasilinear Utility: $u(x, y) = v(x) + y$ for any continuous $v : \mathbb{R} \rightarrow \mathbb{R}$

Consumer Choice: Preliminaries

- We will start our study of consumer choice using the framework we have constructed.
- We will build a theory of choice based on preferences/utility functions.
- If consumers show some consistency in the choices they make, they behave **as if** they were maximizing a utility function.
- Suppose we have an individual who must choose between two goods x and y .
- The consumption space is thus $\mathbb{R}_+^2$, i.e. the set of all consumption bundles in the plane.
- The individual, however, is **constrained** by her income. She faces a **budget restriction**.
- Limited income means that this individual needs to choose x and y in such a way that it satisfies the budget restriction while making her utility $u(x, y)$ as large as possible.

The budget constraint

- A budget constraint is a constraint on how much money (income, wealth) an agent can spend on goods.
- Let $I \geq 0$ denote available income. $(x, y) \in \mathbb{R}_+^2$ denotes a consumption bundle.
- We assume that we can observe the prices of the goods (p_x, p_y)
- The **budget set** is then described by

$$\{(x, y) \in \mathbb{R}_+^2 \mid p_x x + p_y y \leq I\}$$

- Expenditures on x : $p_x x$, Expenditures on y : $p_y y$. The sum of both cannot exceed I .
- The **budget line** is: $p_x x + p_y y = I$.
- Slope of the budget line $\frac{dy}{dx} = -\frac{p_x}{p_y}$

Changes in the budget line

- Write the budget line as

$$y = \frac{I}{p_y} - \frac{p_x}{p_y}x$$

- Notice that the intercepts are I/p_y and I/p_x .
- Then it is easy to see what happens when we change either prices or income.

The consumers Utility Maximization Problem (UMP)

- We will model consumer choice **as if** he were maximizing utility.
- Do economists think that consumers actually maximize utility? **NO**.
- However, the model will yield some testable predictions that will actually conform to reality.
- Is this a good model? Depends on what you are analyzing.
- Advantage: wide range of applications under the same fundamental mathematical model.

Definition (Utility Maximization Problem)

Suppose that preferences over bundles $(x, y) \in \mathbb{R}_+^2$ are represented by $u(x, y)$, prices are (p_x, p_y) and income is I

$$\begin{array}{ll} \text{maximize} & u(x, y) \\ \text{subject to} & p_x x + p_y y \leq I \end{array} \quad (\text{UMP})$$

Graphical Analysis

The slopes of the utility function and the budget line need to be equal.
Thus

$$\begin{aligned}-\frac{p_x}{p_y} &= -\frac{u_x}{u_y} \\ \frac{p_x}{p_y} &= \frac{u_x}{u_y} \\ \frac{p_x}{p_y} &= MRS\end{aligned}$$

